

Sample End Semester Question Paper - 2025

Subject: Power Electronics | Code: EE401 | Semester: 4

Branch: EEE | Regulation: R20 (sample) | Time: 3 Hours | Max Marks: 100

This is a SAMPLE question paper generated for Backlog Buddy (demo/testing data).
It is not an official university paper and is provided to demonstrate the platform.

PART A - Short questions (10 x 1 = 10 Marks)

1. Name one power semiconductor device.
2. What is a snubber circuit?
3. Define THD.
4. What is a freewheeling diode?
5. Name two types of choppers.
6. What is a thyristor?
7. Name two types of rectifiers.
8. What is an inverter?
9. Define duty cycle of a chopper.
10. What is an AC voltage controller?

PART B - Long questions (6 x 13 = 78 Marks)

1. Explain the construction and working of an SCR with V-I characteristics.
2. Explain single-phase half-wave and full-wave rectifiers with waveforms and calculations.
3. Explain single-phase full-bridge inverter operation with waveforms.
4. Explain step-up and step-down choppers with waveforms and calculations.
5. Explain single-phase AC voltage controllers: phase control and ON-OFF control.
6. Explain power semiconductor devices: diodes, MOSFETs and IGBTs.